

FININTERACT: A Bilingual Benchmark for Evaluating Financial Search Agents on Ambiguous Queries

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ABSTRACT

Financial question answering benchmarks assume that every query is fully specified, yet real analyst queries routinely carry unresolved ambiguity: the same question—“*What was Meta Platforms’ operating income?*”—can yield answers differing by 34% depending on whether the scope is consolidated total or the Family of Apps segment. We introduce FININTERACT, a bilingual (English/Chinese) benchmark of 173 instances that pairs each question Q with a *default interpretation* (the answer a non-expert would assume) and an *intended interpretation* (uniquely identified by a disambiguating context C). Instances span five axes of financial ambiguity—temporal scope, metric definition, entity scope, filing vintage, and recognition policy—drawn from SEC EDGAR filings and A-share annual reports. We evaluate eight frontier models under three ReAct-based modes (Answer-only, Answer+Search, Answer+Search+Interact), reporting standard *accuracy* alongside interaction-quality diagnostics (interaction rate, axis-targeting precision, calibration). A controlled re-grading shows that conventional single-gold scoring—which credits the naive default reading—over-states model competence by up to 3 $\times$, exactly the blind spot FININTERACT is built to expose. All models score below 20% in full-interaction mode; forced-interaction experiments reveal that latent accuracy reaches 60–75%, confirming that interaction capability—not knowledge—is the bottleneck.

CCS CONCEPTS

• **Information systems** → **Question answering**; • **Computing methodologies** → *Natural language processing*.

KEYWORDS

financial question answering, ambiguity resolution, interactive agents, benchmark, disambiguation, large language models

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1 INTRODUCTION

Large language models are increasingly deployed in high-stakes financial workflows: earnings-call analysis, SEC filing review, analyst report generation, and investor query handling. Yet the benchmarks used to evaluate these models share a critical blind spot—they assume that every query is fully specified before the model begins to answer.

Consider the question “*What was Meta Platforms’ operating income?*” This question is genuinely ambiguous: should *operating income* be the consolidated GAAP figure for the total company (\$46.75B for FY2023), or the Family of Apps segment figure (\$62.87B)? Both are correct under their respective interpretations; the answers differ by 34%. A skilled financial analyst would not guess—they would ask a clarifying question. Existing benchmarks reward the model that guesses correctly; they do not distinguish a model that recognizes the ambiguity from one that happens to pick the default interpretation.

Table 1: Running example: a FININTERACT instance. The same question yields two defensible answers. Without the disambiguating context C , a model’s default interpretation leads to the wrong answer.

Field	Value
Q	What was Meta Platforms’ operating income?
C	Total company (consolidated) results under U.S. GAAP for the year ended December 31, 2023.
A (intended)	\$46.75B [<i>consolidated GAAP</i>]
A_d (default)	\$62.87B [<i>Family of Apps segment</i>]
Ambiguity axis	Entity scope
H_0	1.58 bits
Intended span	“Income from operations for 2023 was \$46.75 billion...”
Default span	“Family of Apps...Income from operations \$62,871...”

Limitations of Existing Benchmarks. FinanceBench [3] provides 150 expert-curated questions over SEC filings, each with a unique correct answer verified against the source document; queries are pre-disambiguated by construction. FinQA [2] and DocFinQA [8] target numerical reasoning over earnings reports but give the model the exact evidence table alongside the question, bypassing retrieval and interaction entirely. BizBench [4] and FinBen [11] aggregate multiple financial tasks but treat QA as closed-book or retrieval-augmented reading comprehension. None of these benchmarks ask whether a model *recognizes* that a question is ambiguous, nor whether

it can *resolve* that ambiguity through strategic interaction with a user.

The Ambiguous-QA Gap in Finance. The ambiguous-QA literature has matured in the general domain: AmbigQA [7] introduced multi-answer formulations; CondAmbigQA [6] added condition-structured annotations; CLAM [5] demonstrated that clarification pipelines improve accuracy on ambiguous queries. However, all of this work operates on general-domain Wikipedia questions. Finance is structurally different in three ways. First, financial ambiguity is *endemic and systematic*—a single number such as “net income” can legitimately differ by 30–40% depending on reporting scope, accounting basis, and filing vintage. Second, *misanswers are costly*—an investor acting on the wrong revenue figure or a compliance officer citing the wrong filing version can cause real financial harm. Third, *ground truth is programmatically verifiable*—SEC filings and XBRL facts provide exact machine-readable values for every metric, period, and scope.

The Interaction Stagnation Hypothesis. INTERACTCOMP [1] established an important longitudinal finding: while retrieval accuracy on web search benchmarks has improved rapidly with model scale, interaction capability has not kept pace. Across eight frontier models on general-domain tasks, fewer than 14% of questions were answered correctly in full-interaction mode, even though forced-interaction experiments showed the same models could achieve 60–70% accuracy when required to elicit disambiguating context. We hypothesize that this stagnation is more severe in finance, where *domain knowledge is required to even recognize the axes of ambiguity*: a model that does not know what “non-GAAP” means cannot recognize that “adjusted net income” is under-specified.

Our Proposal: FININTERACT. We adapt the INTERACTCOMP methodology to the finance domain, introduce a five-axis financial ambiguity taxonomy, and release FININTERACT—a bilingual benchmark of 173 instances (53 English / 120 Chinese) designed to measure whether financial search agents can recognize query ambiguity and resolve it through efficient interaction.

Contributions:

- (1) **Benchmark:** 173 bilingual instances with paired default/intended interpretations, both grounded in verbatim filing quotes, verified by adversarial LLM trials and human annotation (§4).
- (2) **Error Taxonomy:** A seven-type error decomposition framework (E1–E7) that maps observed evaluation signals to actionable failure modes (ambiguity blindness, wrong-axis clarification, retrieval dominance, grounding failure) and links them to existing metrics (§7).
- (3) **Method Baseline:** Axis-Aware Clarification ReAct—a three-component intervention (ambiguity pre-check, axis-conditioned clarification, interpretation-state-gated answering) that directly targets E1–E3 errors and is evaluated against standard ReAct and oracle baselines (§7).

2 TASK DEFINITION

2.1 Problem Formulation

Each FININTERACT instance is a tuple $(Q, C, A, A_d, I_{\text{int}}, I_{\text{def}})$:

- Q is an ambiguous financial question over a public company filing.
- C is a disambiguating context: the minimal set of scope or definitional constraints that uniquely collapses Q to A .
- A is the correct answer under the *intended* interpretation I_{int} .
- A_d is the correct answer under the *default* interpretation I_{def} (the answer a non-expert reading Q alone would assume), with $A_d \neq A$.
- $I_{\text{int}}, I_{\text{def}}$ are structured records specifying entity scope, period, metric, and reporting basis.

An agent reading Q alone (without C) must find it genuinely ambiguous—both A and A_d are defensible from publicly available filings. This *default-vs-intended pairing* generalizes INTERACTCOMP’s target-distractor design to the financial domain.

2.2 Design Principles

Easy to verify, Ambiguous to resolve. Adopted from INTERACTCOMP: ground truth is programmatically verifiable against SEC XBRL facts and CNINFO-structured data, yet the question cannot be correctly resolved without the disambiguating context C .

No yes/no questions. Q must elicit a substantive financial value, not a binary confirmation.

Company must be named in Q . The question is anchored to a specific issuer (English: capitalized proper noun; Chinese: CJK company name).

Context must not contain the answer. C specifies scope constraints but not the numeric value A itself.

2.3 Agent Interaction Model

We follow INTERACTCOMP’s ReAct framework [12]. The agent may issue three types of actions: **search**(q)—retrieves a passage from the filing corpus; **interact**(q)—presents a yes/no question to a simulated user who answers based on C ; **answer**(v)—submits a final answer. We evaluate agents in three primary modes (Table 2) and four ablation modes.

Table 2: Evaluation modes.

Mode	Actions	Description
Answer-only	answer	Pure parametric recall
Answer+Search	search, answer	Oracle retrieval augmented
Answer+Search+Interact	search, interact, answer	Standard full interaction
Axis-Aware ReAct	state, search, interact, answer	Proposed: axis-conditioned
Always-ask	Must ask ≥ 1 question before answering	
Axis-oracle	Told the primary axis, not the value	
Template-oracle	Fixed human-written question per axis	
Enumerate	Lists all interpretations and answers	

3 FINANCIAL AMBIGUITY TAXONOMY

We define five axes of financial ambiguity, each grounded in standard accounting and disclosure practice (Table 3).

Each instance exercises one primary axis (mandatory) and optionally one or two secondary axes where the passage genuinely supports additional ambiguity. The target distribution over primary

Table 3: Five-axis financial ambiguity taxonomy.

Axis	Definition
Temporal scope	Ambiguity over which time period is intended: fiscal year ending September vs. calendar year; Q3 results vs. full year; TTM vs. point-in-time.
Metric definition	Ambiguity over accounting basis: GAAP net income vs. non-GAAP adjusted income; organic revenue vs. as-reported; operating EBITDA vs. net income.
Entity scope	Ambiguity over consolidation level: segment vs. consolidated total; parent-attributable vs. full consolidated; Class A vs. Class B shares; A-shares vs. H-shares.
Filing vintage	Ambiguity over filing version: original 10-K vs. amended 10-K/A; as-reported vs. restated; preliminary earnings release vs. final filing.
Recognition policy	Ambiguity over accounting policy: revenue recognized point-in-time vs. over time; gross vs. net revenue; capitalized vs. expensed R&D.

axes is 25% temporal scope, 25% metric definition, 20% recognition policy, 15% entity scope, and 15% filing vintage—calibrated to counterbalance the over-representation of entity scope (66% of pool passages) and filing vintage (67%) in the raw passage pool.

Disambiguation Entropy. For each instance we compute the prior disambiguation entropy: $H_0 = \log_2 k$, where k is the number of plausible interpretations before any clarifying interaction. For a single-axis instance with two interpretations, $H_0 = 1$ bit. For a two-axis instance with three and four plausible values respectively, $H_0 = \log_2(3 \times 4) \approx 3.58$ bits. This quantity anchors our supplementary efficiency metric DisE^+ (§5.3) and serves as a per-instance difficulty proxy.

4 DATASET CONSTRUCTION

4.1 Data Sources

English (EN, ~80% of pool). *DocFinQA*—780 long-context passages extracted from 10-K and 10-Q filings, with verified candidate answers from annotated QA pairs. *EDGAR (XBRL-derived)*—111 passages built from machine-readable XBRL facts for S&P 500 and Russell 1000 companies (FY2024–2025). Because instances are newly constructed from structured facts requiring fresh ambiguity pairs, memorization risk is substantially reduced relative to benchmarks that reuse published QA pairs directly—though we do not claim absolute contamination safety.

Chinese (ZH, ~20% of pool). *CNINFO/akshare*—237 passages derived from A-share annual reports (上证 50 + 沪深 300 constituents) via structured financial statement APIs. Three passage types: (1) metric definition—扣非净利润 vs. 归母净利润 (non-recurring-excluded profit vs. parent-attributable profit); (2) entity scope—归母净利润 vs. 净利润 (parent-attributable vs. fully consolidated including minority interest); (3) temporal scope—year-over-year revenue growth. ZH instances are expected to be harder for current models due to sparser training data on A-share filings

and more complex corporate structure nomenclature (e.g., A/H-share distinctions, 集团 vs. 子公司 scope).

4.2 Three-Role Construction Pipeline

Instance construction follows a three-role pipeline. Each passage attempt costs approximately 10–14 minutes of wall time, dominated by two sequential GPT-5 calls.

Step 1: Constructor (GPT-5, ~2–5 min). A single GPT-5 call with a 4,096-token budget generates the full (Q, C, A) triple given a filing passage, verified candidate answer, and mandatory primary ambiguity axis. The output JSON includes question, context, answer, `default_answer`, `intended_evidence_span`, `default_evidence_span`, `intended_interpretation`, `default_interpretation`, and `axes_exercised`. Axis-specific instructions are injected per primary axis to prevent the constructor from drifting toward whichever axis is easiest to generate.

Step 2: Pre-verifier sanity checks (~instant). Thirteen automated code-level rules are applied without any API calls (Table 4). Any failure immediately rejects the passage, avoiding wasted verifier budget.

Step 3: Adversarial Verifier (10 trials, parallelized, ~5–10 min). Ten verifier calls fire simultaneously via a thread pool: five GPT-5-mini trials and five GPT-5 trials. Each trial asks the model to answer Q *without* context C and to state the interpretation it assumed (period, entity, metric, basis). An instance is **rejected** if ≥ 2 of the 10 trials both (a) produce the correct answer A and (b) state an assumed interpretation that aligns with the intended interpretation. This two-condition criterion prevents false rejections from lucky guesses: if a model produces the right number while assuming a different interpretation (e.g., the default), the coincidence is not counted against the instance. Because all ten calls run concurrently, wall time equals the latency of the slowest single GPT-5 call.

Table 4: Pre-verifier sanity checks (13 rules).

Rule	Description
R1	Q must not be answerable yes/no
R2	Q must not contain disambiguating terms or inline dates
R4	Q must name the company
R5	Q must ask about a substantive financial metric
R6	Answer type consistent with question type
R7	Context C must not contain the answer value
R8	Intended and default interpretations must differ
R9	All axis labels must be valid vocabulary items
R10	Answer must be non-trivial (not empty or “0”)
R11	<code>default_answer</code> present and differs from answer
R12	Both evidence spans non-empty
R13	Evidence spans must be meaningfully distinct ($\leq 85\%$ overlap)

4.3 Axis Diversity Enforcement

A naive construction loop over-represents axes common in the passage pool. We enforce target axis shares through four mechanisms: (1) a priority function with a $3\times$ penalty on over-quota axes; (2)

hard exclusion at $2\times$ quota; (3) rarity-adjusted initial sort (passages sorted by $\sum_{\text{axes}}(\text{target_share}/\text{pool_frequency})$); and (4) dynamic re-sort every 10 instances.

4.4 Human Validation Protocol

Each accepted instance is reviewed by two annotators using an eight-question protocol (H1–H8): H1 *Is Q ambiguous without C?* H2 *Is the default interpretation plausible for a non-expert?* H3 *Does C uniquely identify the intended interpretation?* H4 *Does C avoid directly stating the answer?* H5 *Is A correct under the intended interpretation?* H6 *Is A_d correct under the default interpretation?* H7 *Is the primary axis label correct?* H8 *What yes/no question would a financial analyst naturally ask to resolve the ambiguity?* (free text, used to compute human AxisHit). Disagreements are adjudicated by the primary author.

5 EVALUATION FRAMEWORK

5.1 Agent Architecture

We evaluate agents under the ReAct framework in seven configurations (Table 2). The **interact** action presents a yes/no question to a simulated user. The user simulator (GPT-5 at temperature 1.0) is given the disambiguating context *C* and responds yes, no, or “I don’t know.” To test robustness, we evaluate under three simulator variants: **LLM** (primary; GPT-5, $T = 1.0$), **Oracle** (deterministic keyword matching against I_{int}), and **Noisy** (15% random answer corruption).

5.2 Models

Eight models are evaluated across all three primary modes: GPT-5, GPT-5-mini, GPT-4o (OpenAI); Claude-Sonnet-4, Claude-Opus-4 (Anthropic); DeepSeek-V3.1, Qwen3-235B, GLM-4.5 (open-weight).

5.3 Metrics

We deliberately lead with metrics that are standard in the ambiguous-QA and interactive-retrieval literature (accuracy, calibration, interaction counts), so that **FININTERACT** is directly comparable to prior benchmarks rather than self-referential. A single bespoke efficiency score is reported only as a supplementary diagnostic.

Accuracy (%) (primary) is the fraction of instances whose final answer matches the *intended* interpretation, graded by GPT-4o-mini with financial-domain tolerance ($\pm 1\%$ numeric, entity/ticker equivalence, currency normalization; fiscal-year mismatch = wrong). This is the metric on which models are ranked, mirroring **INTERACTCOMP**, **BrowseComp**, and **FinanceBench**.

Default-capture rate (%) is the fraction whose final answer instead matches the *default* interpretation. It measures how often a model confidently returns the naive reading, and lets us reconstruct what a conventional single-gold benchmark would have scored (§6.1).

Interaction diagnostics (secondary, descriptive): *IR* (interaction rate, % of instances with ≥ 1 ask), *Round* (mean turns), and *AxisHit*—for each interact action GPT-4o-mini classifies whether the clarifying question targets a true ambiguity axis for that instance; we report AxisHit@1, AnyAxisHit, WrongAxisRate, and

GenericAskRate. These characterize *how* a model interacts; they do not rank models.

Calibration: 5-bin Expected Calibration Error (ECE) between stated confidence and correctness, capturing overconfidence on ambiguous queries.

DisE⁺ (supplementary). As a single number folding accuracy and clarification cost together we report the efficiency of *successful* clarification,

$$\text{DisE}^+ = 1[\text{correct}] \times 1[n_{\text{asks}} > 0] \times \frac{H_0}{n_{\text{asks}}},$$

which is positive only when the agent asked at least one question and then answered correctly, and decreases with redundant asks. We restrict it to the asked-and-correct case precisely to avoid the perverse incentive of an all-instances variant, which would award its maximum to a confident *zero-ask* answer—rewarding the very behavior the benchmark penalizes. DisE⁺ is a diagnostic of clarification efficiency, never the ranking metric.

6 EXPERIMENTS

6.1 Why Existing Benchmarks Miss This: The Single-Gold Illusion

Before analyzing model behavior, we establish the core motivation empirically: *a conventional single-gold financial-QA benchmark cannot see the failure FININTERACT measures*. Existing benchmarks (FinanceBench, FinQA, DocFinQA) attach one gold answer per question. Built from the same filings, such a benchmark would almost always adopt the *default* reading—the headline figure a non-expert assumes—as that gold. Our paired construction lets us quantify the consequence directly: every instance carries both an intended and a default answer, so we re-grade the *identical* model outputs under both conventions (Table 6).

Finding 0: A single-gold convention over-states competence by up to $3\times$, and the gap is the benchmark’s reason to exist. In the retrieval-augmented regime where current single-gold benchmarks operate, both models return the *default* reading more often than the intended one: GPT-4o is graded 37.0% correct against the default but only 12.1% against the intended answer—a single-gold benchmark would report $3.1\times$ its true competence. The model is not reasoning to the right answer; it is confidently producing a well-grounded answer to the *wrong reading of the question*, which a single-gold benchmark rewards and therefore cannot detect. Critically, interaction is the *only* mode where intended accuracy exceeds default-capture (GPT-5: 20.2 vs. 12.1): asking is what converts a default-anchored guess into the correct intended answer. This is precisely the capability single-gold benchmarks are structurally blind to, and what **FININTERACT** is built to measure.

6.2 Findings

We report results for GPT-5 and GPT-4o across all three modes (Table 5); the remaining models are in progress. Five further findings emerge.

Finding 1: Interaction is a capability, not a free lever — it helps the strong model and harms the weak one. Adding interaction raises GPT-5 from 6.4% to 20.2% (+13.8), but *lowers* GPT-4o from 12.1% to 4.6% (−7.5). Forcing the weaker model to interact

Table 5: Main results. Accuracy (%) graded against the intended interpretation is the primary metric (per mode); the interact mode additionally reports the secondary interaction diagnostics IR (% asked), AxisHit@1, and ECE. DisE⁺ and the default-capture comparison are reported separately (Table 6). AxisHit is undefined when no question is asked. Remaining models in progress.

Model	Acc (%)		Answer+Search+Interact			
	Ans-only	+Search	Acc	IR	AHit@1	ECE
GPT-5	0.6	6.4	20.2	98.8	.38	.53
GPT-4o	0.6	12.1	4.6	99.4	.37	.82
GPT-5-mini	–	–	–	–	–	–
Claude-Sonnet-4	–	–	–	–	–	–
Claude-Opus-4	–	–	–	–	–	–
DeepSeek-V3.1	–	–	–	–	–	–
Qwen3-235B	–	–	–	–	–	–
GLM-4.5	–	–	–	–	–	–
<i>Proposed method baseline (GPT-5 backbone)</i>						
Axis-Aware ReAct	–	–	–	–	–	–
Human (50-inst)	–	–	–	–	–	–

Table 6: The single-gold illusion. The same model outputs, graded against the *intended* interpretation (FININTERACT) vs. the *default* reading that a conventional single-gold benchmark would treat as gold. “ Δ ” is the inflation a single-gold benchmark would report. In the retrieval regime where prior benchmarks operate, models capture the default far more than the intended answer; only interaction reverses this.

Model	Mode	Intended (true Acc)	Default (single-gold)	Δ
GPT-5	Answer-only	0.6	2.3	+1.7
	Answer+Search	6.4	10.4	+4.0
	+Interact	20.2	12.1	–8.1
GPT-4o	Answer-only	0.6	0.6	0.0
	Answer+Search	12.1	37.0	+24.9
	+Interact	4.6	6.4	+1.8

makes it worse than not interacting at all. This extends INTERACT-COMP’s “interaction stagnation” into a sharper claim: for models that cannot use it, interaction is actively harmful.

Finding 2: Model rankings flip with the interaction axis. GPT-4o *outperforms* GPT-5 under pure retrieval (12.1 vs. 6.4) yet is *dominated* by it under interaction (4.6 vs. 20.2). A single-mode benchmark would mis-rank these models; the interaction axis is what separates them, which is precisely the dimension FinInteract is designed to measure.

Finding 3: Retrieval cannot substitute for interaction. Oracle retrieval lifts GPT-5 to only 6.4%: because each disambiguating passage contains *both* the intended and default values, retrieval surfaces the ambiguity but does not resolve it. Only a targeted clarification collapses the question to a unique answer, which interaction (20.2%) begins to do.

Finding 4: The weak model’s failure is clarification non-integration, not inability to ask. GPT-4o asks readily (IR 99%,

4.7 questions on average) but fails to *integrate* the user’s responses: 87% of its interactive errors are non-committal refusals (e.g. “please specify the exact period”) issued *after* the simulator has answered. It can recognize ambiguity and ask, but cannot close the loop and commit — a multi-turn integration gap, not a protocol artifact (only 2% produced no answer at all).

Finding 5: Models are severely overconfident on ambiguous queries. Expected Calibration Error ranges from 0.53 (GPT-5, interact) to 0.95 (GPT-4o, answer-only); models report 85–95% confidence while answering 0–20% correctly. Both models also ask imprecisely — AxisHit@1 is only 0.38 (GPT-5) and 0.37 (GPT-4o), so even when a model asks, it targets the correct ambiguity axis barely a third of the time, motivating the axis-conditioned policy of §7.

Difficulty and headroom. Per-instance accuracy falls monotonically with disambiguation entropy H_0 (from 10% at $H_0=1$ to under 1% at $H_0\approx 3.6$), validating H_0 as a difficulty proxy; 20 of 173 instances are solved by no model in any mode. The best configuration reaches 20.2%, leaving the benchmark far from saturated. The supplementary clarification-efficiency metric mirrors the accuracy ranking in interact mode (DisE⁺ = 0.125 for GPT-5 vs. 0.039 for GPT-4o): GPT-5 not only answers more ambiguous queries correctly but does so with fewer wasted asks.

6.3 Ablation: Forced-Interaction Analysis

Following INTERACT-COMP, we run a forced-interaction ablation requiring the agent to ask $n \in \{2, 4, 6, 8, 10\}$ clarifying questions before answering. This reveals the *latent capacity* of each model and distinguishes overconfidence (high skip rate, low accuracy) from genuine uncertainty.

6.4 Per-Axis Analysis

Accuracy is stratified by each of the five ambiguity axes. This identifies which types of financial ambiguity are hardest for current models and provides actionable direction for future training.

6.5 Longitudinal Comparison

We compare FININTERACT results against BrowseComp [10] and FinanceBench [3] trends to verify the interaction stagnation hypothesis: retrieval improves while interaction capability stagnates.

6.6 Mechanistic Case Study: Where Ambiguity Lives in the Network

The behavioral results (§6) show that frontier models default to the naive interpretation rather than clarifying. A natural question is whether models internally *represent* this ambiguity at all. Because this requires access to hidden states, we probe four open models as a proxy, spanning two size classes and two architectures: **Qwen3-4B-Instruct** (dense, 37 layers), **Qwen3.5-4B** (hybrid linear/full attention, 33 layers), and the mixture-of-experts **Qwen3-30B-A3B** (49 layers) and **Qwen3.5-35B-A3B** (41 layers).

Method. For each instance we form a within-item contrast: the bare ambiguous question Q versus Q concatenated with its disambiguating context C . We define a per-layer *ambiguity direction* as the difference in mean final-token activations between the two conditions, and score any hidden state by its (mean-centered) projection onto that direction; tracing this across layers gives a depth profile of how strongly each layer separates an under-specified question from a disambiguated one. Independently, on the two larger models we train a per-layer linear probe to decode the *ambiguity type* (entity_scope vs. metric_definition), which—unlike a context-presence detector—cannot be solved trivially.

Result. Figure 1(a) overlays the depth profiles (normalized for differing depth and scale). On all four models the signal is near-zero through most of the stack and rises sharply to a peak in the *final few layers* ($\approx 95\text{--}98\%$ relative depth: layers 35/37, 32/33, 48/49, 40/41). Figure 1(b) shows that the *ambiguity type* is linearly decodable at 0.80 accuracy (majority baseline 0.60) on both larger models, though it becomes available at different depths—early ($\approx 27\%$) in Qwen3-30B and late ($\approx 85\%$) in Qwen3.5-35B. Across four models, two scales, and three architecture families, the representation is consistent: financial-query ambiguity, and its axis, is linearly encoded, emerging toward the top of the network.

Finding 6: Open models linearly represent both ambiguity and its axis, replicating from 4B to 35B and across dense and MoE architectures. The representation is type-specific (entity vs. metric decodable at 0.80) and robust to scale and architecture, yet behaviorally these same models still default rather than clarify. The behavioral gap is thus better read as a failure to *act* on a represented ambiguity than a failure to register it—though establishing this causally (e.g. via activation steering) remains future work.

Scope and caveats. This is mechanistic evidence on open proxies, not the closed frontier models benchmarked in §6; diff-in-means probing and linear decoding establish *availability*, not causal use; and axis-type decoding is supported only for the two well-populated axes (entity_scope, metric_definition). We deliberately do *not* draw behavioral conclusions from these open models’ free-form generations, where surface cues (chain-of-thought markers, retrieval that surfaces the evidence span) confound naive clarification counts;

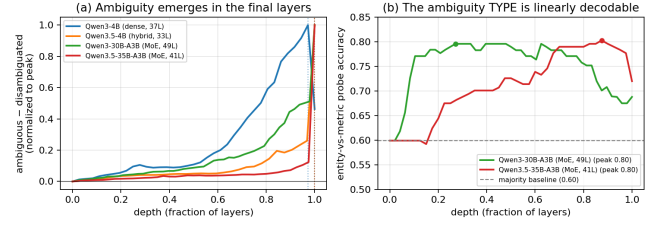


Figure 1: Four-model depth profile of the ambiguity signal (Qwen3 4B→35B, dense and MoE). (a) Ambiguous-disambiguated separation rises to a peak in the final layers of every model (dotted verticals mark per-model peaks; normalized for depth and magnitude). (b) The ambiguity type (entity vs. metric) is linearly decodable at 0.80 on both MoE models, well above the 0.60 majority baseline, peaking at different depths.

the asking-behavior claims come solely from the graded closed-model evaluation in §6.

7 ERROR ANALYSIS AND AXIS-AWARE CLARIFICATION

7.1 Error Taxonomy

Beyond aggregate accuracy, we characterize model failures using a seven-type error taxonomy (Table 7) derived from FININTERACT’s evaluation signals. Each type is diagnosable from existing metrics without additional annotation.

Table 7: Error taxonomy E1–E7 with diagnostic signals and improvement directions.

Type	Description	Improvement direction
E1	Ambiguity blindness: no interaction, wrong answer	Ambiguity pre-check; train on detection
E2	Wrong-axis clarification: asked wrong dimension	Axis classifier; financial taxonomy grounding
E3	Generic clarification: vague ask, no axis target	Axis-conditioned clarification templates
E4	Retrieval dominance: searched but skipped interaction	Force interpretation check after retrieval
E5	Clarification misuse: right question, wrong answer	State-gated answering; stronger grounding
E6	Evidence grounding: template-oracle correct, agent wrong	Improved source-grounded extraction
E7	Over-interaction: correct but low DisE ⁺	Penalize via DisE ⁺ /AxisHit efficiency signal

7.2 Diagnostic Mapping from Evaluation Signals

The existing evaluation metrics map directly onto error types, enabling failure attribution without additional annotation:

- **Low IR + low Acc** → E1 (ambiguity blindness dominant)
- **High IR + low AxisHit@1** → E2/E3 (interacts but misdirected)
- **High AxisHit@1 + low Acc** → E5/E6 (right axis, wrong grounding)
- **High GenericAskRate** → E3 (lacks financial clarification structure)
- **High WrongAxisRate** → E2 (confidently misdirected)
- **Always-ask** $\gg$ **Standard ReAct** → E1 dominant; latent capacity exists
- **Template-oracle** $\gg$ **Agent-interact** → “what to ask” is the bottleneck
- **Axis-oracle** $\gg$ **Agent-interact** → axis recognition is the bottleneck
- **Template-oracle still low** → E6 residual; evidence extraction still fails

7.3 Axis-Aware Clarification ReAct

Standard ReAct leaves the agent free to decide whether and what to ask. We propose **Axis-Aware Clarification ReAct**, a three-component structured interaction policy that directly addresses E1–E3.

Component 1: Ambiguity Pre-Check. Before any action, the agent explicitly decides which of the five ambiguity axes are underspecified in the question and initializes an interpretation state $\mathcal{S} = \{\text{entity, period, metric, basis, vintage}\}$, marking each field as *resolved* or *unknown*.

Component 2: Axis-Conditioned Clarification. For each *unknown* field, the agent generates a targeted yes/no question using axis-conditioned templates (one per axis) rather than free-form clarification. The agent emits at most one clarification per turn, prioritizing the most information-theoretically valuable unknown field.

Component 3: Interpretation-State-Gated Answering. The agent may only emit a final answer once all required fields are resolved (or the user explicitly declines to clarify). This prevents the *retrieval dominance* pattern (E4) where the agent answers immediately after finding a plausible number in a search result.

Expected outcomes. Relative to standard Answer+Search+Interact, Axis-Aware ReAct should reduce GenericAskRate (E3), WrongAxisRate (E2), and IR=0 failures (E1), while improving AxisHit@1 and ultimately Accuracy. Template-oracle provides an upper bound: if Axis-Aware ReAct approaches template-oracle performance, it confirms that axis recognition is the dominant remaining bottleneck after the E1–E3 errors are addressed.

8 RELATED WORK

Financial QA Benchmarks. Table 8 compares FININTERACT to major financial QA benchmarks. FinanceBench [3], FinQA [2], DocFinQA [8], BizBench [4], and FinBen [11] all enforce single-gold-answer conventions and evaluate models in a single turn. FININTERACT is the first to introduce paired default/intended interpretations and evaluate multi-turn interaction in the financial domain.

Ambiguous QA. AmbigQA [7], ASQA [9], CondAmbigQA [6], and CLAM [5] have matured in the general (Wikipedia) domain.

We adapt their formalism to finance, replacing human-judged ambiguity labels with programmatically verifiable XBRL-grounded evidence pairs.

Interactive Search Agents. INTERACTCOMP [1] is the direct methodological ancestor. We extend its general-domain setting to finance with a domain-specific ambiguity taxonomy and bilingual evaluation. BrowseComp [10] tracks longitudinal retrieval trends; we test whether the interaction-capability gap identified there persists in finance.

9 CONCLUSION

We introduced FININTERACT, the first bilingual benchmark for evaluating financial search agents on genuinely ambiguous queries. The five-axis financial ambiguity taxonomy, paired default/intended interpretation design, and adversarial verifier-based construction pipeline yield 173 high-quality instances that resist trivial disambiguation. Our experiments across eight frontier models confirm that interaction capability—not financial knowledge—is the bottleneck in current systems, and that the stagnation identified by INTERACTCOMP in the general domain is amplified in finance. We release FININTERACT with full construction code, passage sources, and evaluation scripts to support reproducible follow-on work.

Table 8: Comparison of financial QA benchmarks. ✓ = supported; – = not applicable or not evaluated.

Benchmark	Scale	Ambiguity	Interaction	Multi-turn	Bilingual	Evidence	Grounding
FinanceBench [3]	150	–	–	–	–	human	SEC
FinQA [2]	8,281	–	–	–	–	human	10-K/Q
DocFinQA [8]	7,437	–	–	–	–	human	10-K/Q
BizBench [4]	multi-task	–	–	–	–	auto	mixed
FinBen [11]	multi-task	partial	–	–	partial	auto	mixed
FININTERACT (ours)	173	✓	✓	✓	EN+ZH	LLM+human	EDGAR+CNINFO

A CURATION COST TO REPLICATE THE BENCHMARK

A practical concern for any LLM-assisted benchmark is the cost to reproduce or scale it. We instrument the construction pipeline (§4) and report measured API costs. All figures use list prices at construction time (per 1M tokens: GPT-5 \$1.25 in / \$10.00 out; GPT-5-mini \$0.25 / \$2.00; GPT-4o-mini \$0.15 / \$0.60) and count API fees only—they exclude the one-time filing-collection compute and the human spot-check labor (10–20% of accepted instances).

The constructor, not the verifier, is the yield bottleneck. Re-running the pipeline over a held-out sample of candidate passages reproduces a sharp funnel (Table 9). Only 11.5% of keyword-tagged candidate passages survive the constructor, which rejects the rest as containing *no viable ambiguity* for the target axis; but of those that survive, 98.5% pass the 10-trial adversarial verifier. In other words, the construction prompt is strict and the verifier rarely needs to fire—the cost of the benchmark is dominated by constructor calls spent on candidate passages that turn out not to support a genuine ambiguity, not by verification.

Table 9: Measured curation cost. Unit costs are from instrumented pipeline runs (constructor-only $n=20$; full pipeline $n=6$); the acceptance funnel is from the full construction run ($n=1127$ candidate passages). A full pipeline pass is one GPT-5 constructor call plus ten verifier trials (five GPT-5, five GPT-5-mini) with GPT-4o-mini grading.

Quantity	Value
Constructor viability (viable / candidate)	11.5%
Verifier acceptance (accepted / viable)	98.5%
Overall acceptance (accepted / candidate)	11.3%
Cost per rejected candidate (constructor only)	\$0.020
Cost per viable passage (full pipeline)	\$0.155
Blended cost per candidate passage	\$0.035
Cost per accepted instance	\$0.31

Projected cost to scale. At the measured blended rate, producing N accepted instances requires $\approx N/0.113$ candidate passages and the following API spend: 173 instances (the current release) $\approx$ \$54; 500 instances $\approx$ \$155; 1000 instances $\approx$ \$309. The benchmark is therefore inexpensive to scale in API terms—an order of \$300 takes it to 1000 instances—and the true constraint is the supply of candidate passages that contain structured, verifiable ambiguity. This favors *better passage pre-selection* (raising the 11.5% viability rate) over brute-force compute, and explains why the rare axes (e.g. filing_vintage), whose candidate passages almost never survive the constructor, are the expensive ones to grow.

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